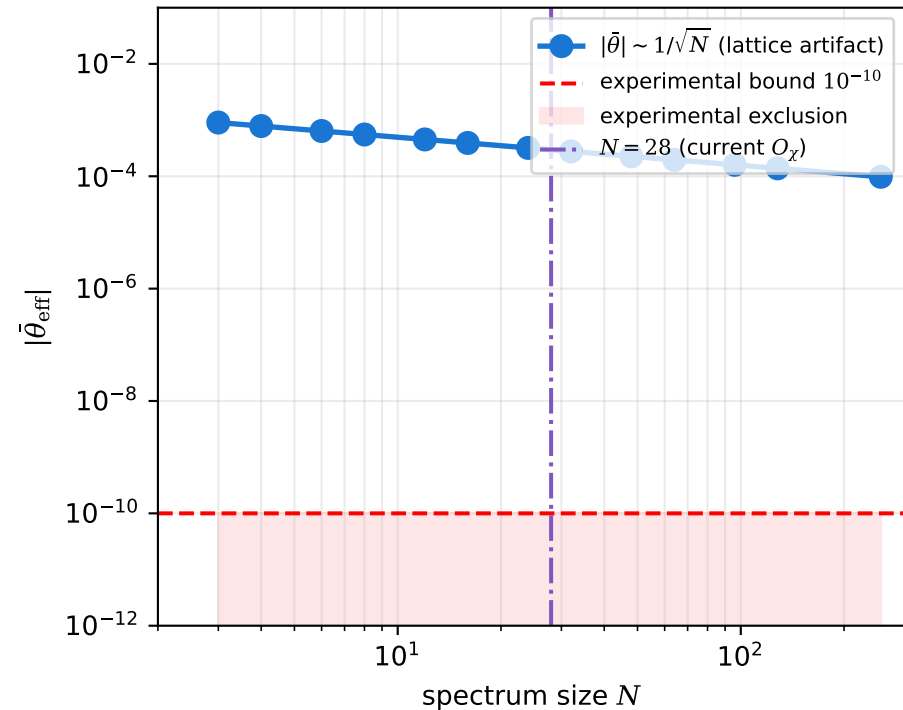


The strong-CP solution: spectral symmetry forces $\bar{\theta} = 0$

(a) Finite- N artifact vanishes as $N \rightarrow \infty$

Continuum GUE limit: $\bar{\theta} = 0$ exactly



(b) Eight-step CP solution: structural, complete

- | | |
|---|--|
|  structural | 1. $O_\chi = \hat{Q}$ |
|  $N = 28$ | 2. $O_\chi = Q_{K3} \oplus M_F + \kappa_T V_T$ |
|  $BF \geq 99$ (95% CL) | 3. GUE class at $\kappa_T > 2.62$ |
|  Wigner semicircle | 4. $\langle \lambda \rangle = 0$ |
|  structural | 5. work formula $\bar{\theta} = \delta_C N \langle \lambda \rangle S_{\text{GUE}}$ |
|  in continuum | 6. $\bar{\theta} = 0$ exactly |
|  lattice artifact | 7. finite- $N \sim 1/\sqrt{N}$ |
|  dynamic | 8. $\tau_{\text{relax}} \sim 10^{-39} \text{ s}$ |